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THE PALAEOMAGNETISM OF OUTCROPPING OCEANIC CRUST ON MACQUARIE ISLAND

By P. WILLIAMSON

(With 3 Tables and 6 Figures)

(MS received 8 August 1978)

ABSTRACT

Macquarie Island offers a rare opportunity to investigate outcropping ocean crust. In this study, palaeomagnetic samples were collected from 32 sites. After frequency demagnetization, 15 of these sites were found to be stable. Stable sites were from lavas, dykes, and gabbros, representing different depths of formation in the oceanic crust.

A successful bedding correction, assuming a simple model for the oceanic crust, was applied to the directions of magnetization. The former assumed dykes were injected vertically, lavas laid down horizontally, and gabbros layered horizontally. However, the palaeomagnetic poles from the centre and south of the island were not conformable at their 95 percent confidence levels. To remedy this, a structural correction was then applied to sites from the south of the island, where the average strike of dykes is 55° different from that in the centre of the island, which parallels regional seafloor-spreading anomalies. This correction involved rotation around a vertical axis, which could not be accounted for in a simple bedding correction. When the average strike of the dykes was brought into parallelism with that of the regional seafloor-spreading anomalies (and presumably that of the fossil spreading ridge), the result was to bring the palaeomagnetic poles from the centre and the south of the island into conformity.

The final pole for Macquarie Island was found to be consistent with a pole of similar age from the Australian Continent. This is consistent with Macquarie Island being formed as part of the Indian plate.

The magnetic properties of the rocks are consistent with seafloor-spreading anomalies originating from the lavas, with a possible deeper contribution from the gabbros. The dolerite dykes which are palaeomagnetically stable appear to have natural remanent magnetizations (NRM) which are too low to contribute significantly to seafloor-spreading anomalies.

INTRODUCTION

Macquarie Island offers a rare opportunity to study oceanic crust in outcrop. Varne & Rubenach (1972) argue from petrological evidence that the island rocks are similar to those formed at accreting plate margins. The work of Williamson & Johnson (1974) suggests that the uplift of oceanic crust forming Macquarie Island and the central part of the Macquarie Ridge Complex is the result of compression at the Indian-Pacific plate boundary. This boundary lies within the Macquarie Ridge Complex which extends as an arcuate system of ridges and troughs from the Macquarie triple junction between the Indian, Pacific and Antarctic plates (shown by Sykes (1970) to be near Lat. 62°S, Long. 160°E) to the Alpine fault at the south of New Zealand. Macquarie Island lies at Lat. 54.5°S, Long. 150°E (Fig. 1).

The interpretation by Williamson (1974) of seafloor-spreading anomalies in the Macquarie Island region suggests that the island rocks were formed at the Indian-Antarctic spreading ridge about 27 m.y. ago. Thus, the palaeomagnetic pole for Macquarie Island should be an estimate of the pole for the Indian plate at

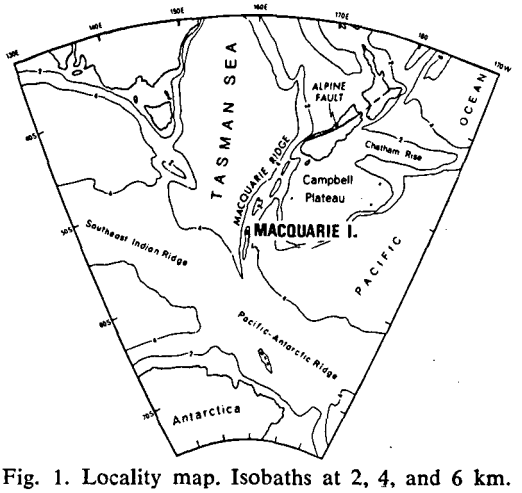


Fig. 1. Locality map. Isobaths at 2, 4, and 6 km.

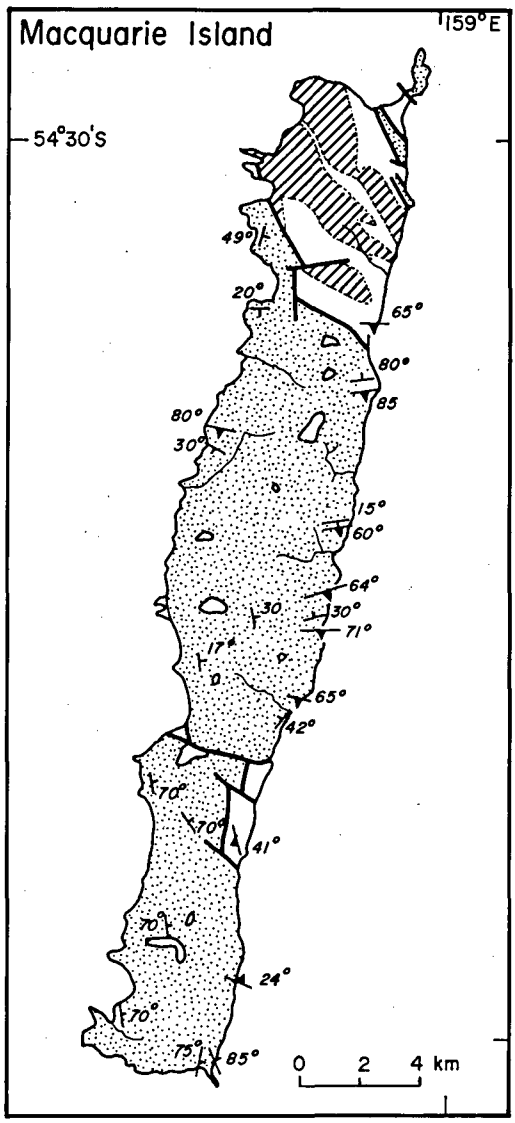


Fig. 2. Generalized geological map of Macquarie Island. Serpentines and gabbros are marked by dashes; extrusive volcanic rocks and associated sediments are marked by dots; dyke swarms are blank. Dips on lavas are shown with a single tick and dips on dykes with a vee. Faulted contacts are shown as heavy lines and gradational or uncertain contacts as dotted lines (after Varne & Rubenach, 1972).

that time. The directions of magnetisation obtained in this study also help to unravel the structural history of the island.

GEOLOGY

Intrusive and extrusive rocks outcrop on Macquarie Island (Fig. 2). The outcrop in the northern quarter of the island consists of metamorphosed tholeiitic lavas, dolerite dykes, gabbros, and serpentinites. In contrast, the outcrop geology of the southern three-quarters of the island is mainly relatively unmetamorphosed lava.

It is possible that metamorphism could have been caused by deformation associated with uplift of the Macquarie Ridge. However, Varne & Rubenach (1972) argue that the type of metamorphism observed is that usually caused by deep burial. Furthermore, the grade of metamorphism of the intrusive rocks, formed at deeper levels in the ocean crust, is greater than that of the lavas, formed at surficial levels. This suggests burial metamorphism, occurring normally in the oceanic crust.

A simple model for the structure of oceanic crust could incorporate horizontal lavas and vertical dykes injected parallel to the spreading ridge. The structure of Macquarie Island is far more complex; the island rocks are heavily faulted and steep, variable dips are observed for the dolerite dykes, and for the layering in the pillow lavas. Nonetheless, in many localities the strikes of lava flows approximately parallel the strikes of the dykes that cut them, and on this evidence Varne & Rubenach (1972) argue that tilting has often occurred about horizontal axes roughly parallel to the strike of the dykes. The predominant strike of the dykes is approximately eastnortheast parallel to the regional strike of seafloor-spreading magnetic anomalies as interpreted by Williamson (1974).

COLLECTION OF SAMPLES

Eighty-six oriented block samples were collected from 32 sites (Fig. 3). Twenty of these sites were located in individual dolerite dykes, nine in basaltic lava flows, and one in each of serpentinite, gabbro, and red sediment.

Because of the normally overcast weather conditions, samples were oriented using a compass and inclinometer. Although the compass bearing was not visibly effected by the magnetism of the rock outcrops, small errors in compass bearing may have resulted. Because of this, and the difficulty in obtaining flat sampling surfaces, orientations of samples may contain small errors. Since this method was also used to measure the strikes and dips of contacts and bedding planes, a similar accuracy can also be expected for those measurements. The measured orientation of bedding planes in pillow lavas is subject to further error which comes from the difficulty in measuring the orientation of individual pillows relative to the general layering of the lava. Added to this could be differences in remanent magnetic

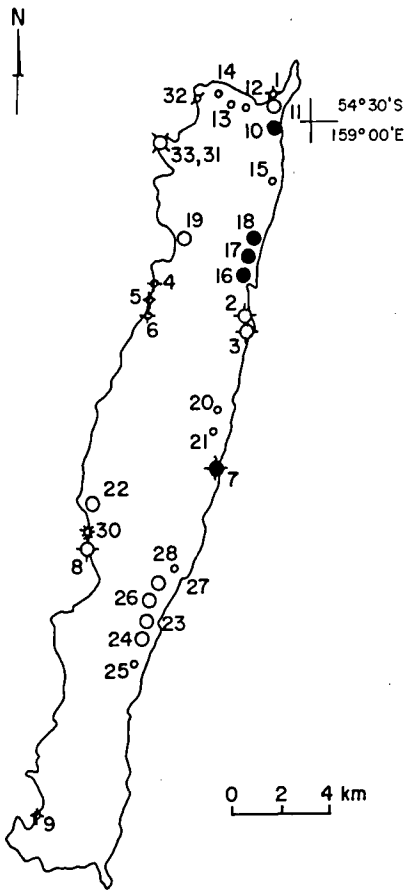


Fig. 3. Location of palaeomagnetic sites on Macquarie Island. Small circles show the location of unstable sites and large circles the location of stable sites. Crosses on the sites refer to lava sites, x's to gabbro sites, the star to the red-sediment site, and the dash to the serpentinite site. For stable sites the normal directions of magnetization are indicated as open circles and reverse directions as closed circles. Site 33 is the gabbro site obtained by Butler (*pers. comm.*, 1974).

directions between the surface and interior of lava pillows. In practice, it was found that the average within site dispersion for stable lava sites was about 9° whereas that for stable dyke sites was about 12°.

These guidelines were followed in measuring orientations of bedding in the field. First, in the case of contacts between intrusive rocks in dyke swarms, it is necessary to know which is the intruding dyke, as the measured orientation of the contact is only valid for the intruding dyke. Thus, the sampling of dyke swarms was carried out with particular attention to the structural relationship. In practice this may not have been crucial since parallel intrusion appeared common. Second, in measuring the orientation of gabbros and serpentinites it was assumed that layering represented nearly horizontal surfaces at the time of their formation, that is, the layering was formed by gravity differentiation. Palaeomagnetic directions determined from these rock types are therefore qualified by this assumption.

LABORATORY MEASUREMENTS

Test specimens for the different localities and rock types were progressively demagnetized in four or more steps up to a maximum of 600oe peak value using the alternative field demagnetizing apparatus described by McElhinny (1966). Initial changes in directions usually occurred in fields less than 100oe. Thereafter vector rotation typically ceased and changes occurred only in intensity. Thus 100oe was taken as the alternating field required to reach an end point as defined by many workers (As & Zijderveld (1958) and McElhinny & Gough (1963)).

Since the sampling density was inadequate for a within-site dispersion test, the criterion used for stability of sites was similar to that used by Larochelle (1968) in judging the stability of samples. In that instance, a sample was considered stable if the angular dispersion between two specimens from the sample after magnetic cleaning was less than 15°. In the present case, because the angular dispersion between samples from the same site would be expected to be greater than between that of specimens from the same sample, an angular dispersion of 30° was used. Thus, in this study, a site is considered stable if the angular dispersion of the samples from the sites after cleaning in a 100oe alternating field is less than 30°. In general, little rotation (less than 20°) of the direction of magnetization occurred during the cleaning of specimens from stable sites.

Fifteen of the 32 sites were found to be stable using the above criterion. Eleven of these represent dykes, and four, lava flows. Because only one of each of the red-sediment and gabbro samples could be reoriented in the laboratory, directions from those sites are not used in this study. Three serpentinite samples could be reoriented but the serpentinite site was not stable under the above criterion. It must be added, however, that no conclusion can be drawn from this study about the general magnetic stability of the serpentinite.

There appears to be some geographical grouping of stable and unstable sites. Dyke sites

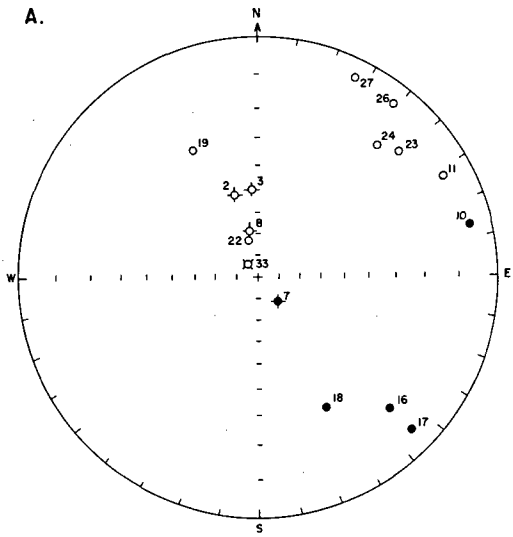


Fig. 4a. *In situ* directions of remanent magnetisation. Open circles indicate normal directions of magnetization and closed circles indicate reversed directions of magnetization. Crosses refer to lava sites, x refers to the gabbro site and plain circles refer to dyke sites. The numbers refer to sites the locations of which were given in Figure 3.

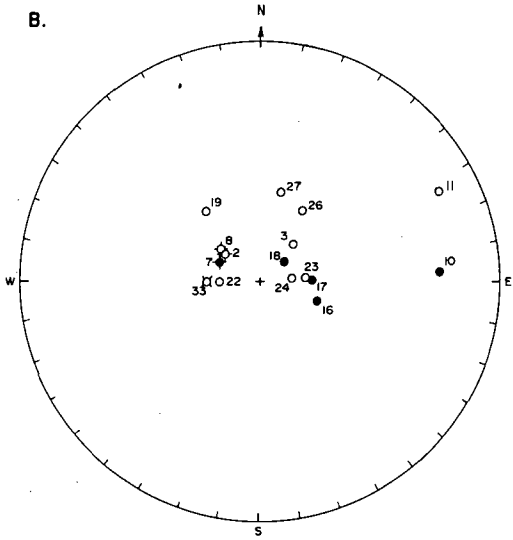


Fig. 4b. Directions of remanent magnetization after bedding correction.

1, 12, 13, 14, and 32 occur adjacent to each other in the north of the island and are unstable. A similar situation also exists for unstable lava sites 4, 5, and 6, stable sites 2, 3, 16, 17, 18, and 19 (lavas and dykes) and stable sites 23, 24, 26, and 27 (dykes).

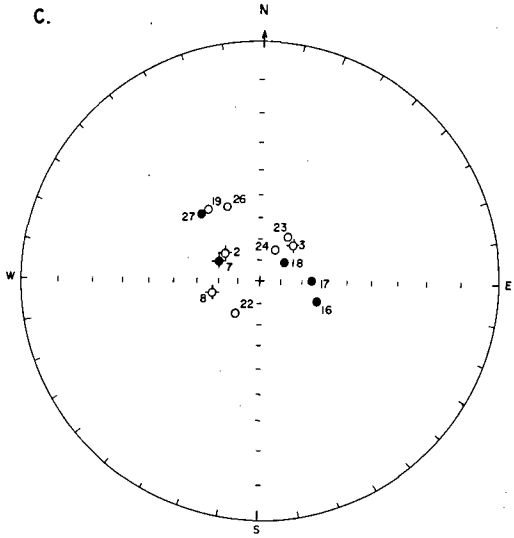


Fig. 4c. Structurally adjusted directions from blocks 2 and 3.

TABLE 1

Direction of Magnetization from Stable Sites

Site	Rock Type	*N*	w.r.t. horizontal *D*	w.r.t. horizontal *I*	w.r.t. bedding *D*	w.r.t. bedding *I*
2	basalt lava	2	345.1	-51.3	303.8	-69.2
3	basalt lava	3	357.2	-49.9	47.5	-65.4
7	basalt lava	2	140.0	74.7	297.5	68.5
8	basalt lava	2	350.8	-69.6	290.2	-59.3
10	dolerite dyke	2	78.0	8.7	88.2	17.8
11	dolerite dyke	3	61.9	-7.4	63.3	-9.5
16	dolerite dyke	3	134.9	13.6	105.6	59.4
17	dolerite dyke	3	134.9	4.8	89.8	63.2
18	dolerite dyke	3	152.8	25.9	57.7	74.3
19	dolerite dyke	3	322.6	-29.3	309.8	-47.6
22	dolerite dyke	3	346.0	-72.9	261.9	-70.4
23	dolerite dyke	3	49.2	-13.5	86.8	-67.5
24	dolerite dyke	2	43.0	-16.9	82.5	-74.5
26	dolerite dyke	3	38.9	-4.5	24.5	-59.4
27	dolerite dyke	2	27.1	-4.2	6.4	-59.2

N = Number of Samples

RESULTS

(i) In situ *directions*

The directions of magnetization with reference to horizontal (Fig. 4a, Table 1) are grouped broadly into three categories. The first is a group of three lavas and two dykes, which show normal magnetization with inclinations ranging from -29° to -73° and declinations between 323° and 357°. The second group of one lava and three dykes, has inclinations ranging from 5° to 75° with declinations between 134° and 152° and probably corresponds to a reverse magnetization to that of the first group. The third group, consisting of six dykes, has inclinations between

5° and -17° and declinations between 27° and 77°, five of which have the same polarity as the Earth's present field and one, the reversed polarity. No stable gabbro site was obtained in the present study. However, Butler (*pers. comm.*, 1974) found in a study of gabbro samples from the north of the island, that the *in situ* direction for his stable gabbro site had inclination -86° and declination 276°. This site thus falls within the first group of the present study.

Comparison between the *in situ* directions of magnetization and the localities of sites (Fig. 3) reveals some consistency in the directions from individual areas. The directions from lava sites 2 and 3 collected at nearby localities plot close together as do those of lava site 8 and dyke site 22 collected from localities within a few kilometres of each other. The same holds true for dyke sites 16, 17, and 18, dyke sites 23, 24, 26, and 27, and dyke sites 10 and 11. This agrees with structural information (Fig. 2), which suggests that in spite of the intense faulting, regional tilts are often fairly constant in both direction and magnitude.

(ii) *Simple bedding corrected directions*

A simple bedding correction was carried out. In this correction, a model for primary oceanic crust was chosen by which lavas and gabbros were deposited horizontally and dykes injected vertically. The *in situ* directions were corrected by rotations about strike, corresponding to returning lavas and gabbros to horizontal and dykes to vertical. The results are shown in Figure 4b, Table I. The application of this correction brought 14 of the 16 directions into closer coincidence and reduced the radius of the circle of 95 percent confidence from 25° to 12° (the latter coefficients were calculated using the normal directions and the opposites of the reversed directions.) It would thus appear that tilting has occurred in most areas after the formation of the rocks.

The simple bedding correction does not correct for rotation of a vertical dyke about a horizontal axis perpendicular to the strike of the dyke. In the remaining two of the above 16 sites (dyke sites 10 and 11) tilting about such axes would be required to bring their directions into reasonable coincidence with the remainder of the corrected directions of magnetization. This implies that the sense of tilting in the extreme north of the island is similar to that in the south. A structural division of the island on direction of tilt would thus be as shown in Figure 5. An apparent pole position was calculated using the directions of

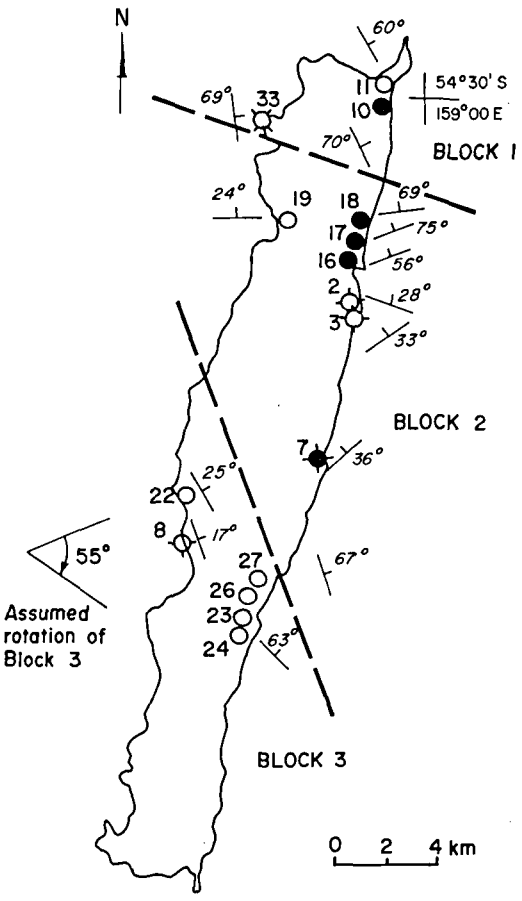


Fig. 5. Structural division of Macquarie Island on the basis of palaeomagnetic results and structural information given in Figure 2. Ten dashed lines refer to approximate boundaries between areas characterised by tilts which are either ENE or WNW. Tilts given are those indicated from the palaeomagnetic results. Only one value is given from sites 23 and 24 and also only one value is given from sites 26 and 27 since both these pairs of sites represent a dyke intruding another dyke and only the orientation of the contact is known.

magnetization except those from dyke sites 10 and 11. The pole was found to be at Lat. 75°S, Long. 18°E (Table 2).

(iii) *Structurally adjusted directions*

On inspection, the palaeomagnetic poles from blocks 2 and 3 (Table 2, Fig. 6) reveal a discrepancy. The poles are in fact inconsistent with each other at their 95 percent confidence levels. Block 1 contains only three sites and could be consistent with either block 2 or 3; it is consequently omitted from this discussion. A further difference between blocks 2 and 3 is also evident from structural information,

TABLE 2

Pole Positions

Rock Unit	Age(m.y.)	South Pole Lat.	South Pole Long.	A95 deg	Reference
1. Block 2 w.r.t. bedding		62°S	41°E	18	
2. Block 3 w.r.t. bedding		75°S	269°E	19	
3. Macquarie Island		75°S	18°E	12	
4. Block 3 after structural adjustment		74°S	48°E	19	
5. Macquarie Island final	27 approx.	68°S	43°E	12	This paper
6. Southeast Queensland	22 - 25	86°S	78°E	12	Robertson (1966), Webb *et al.* (1967)

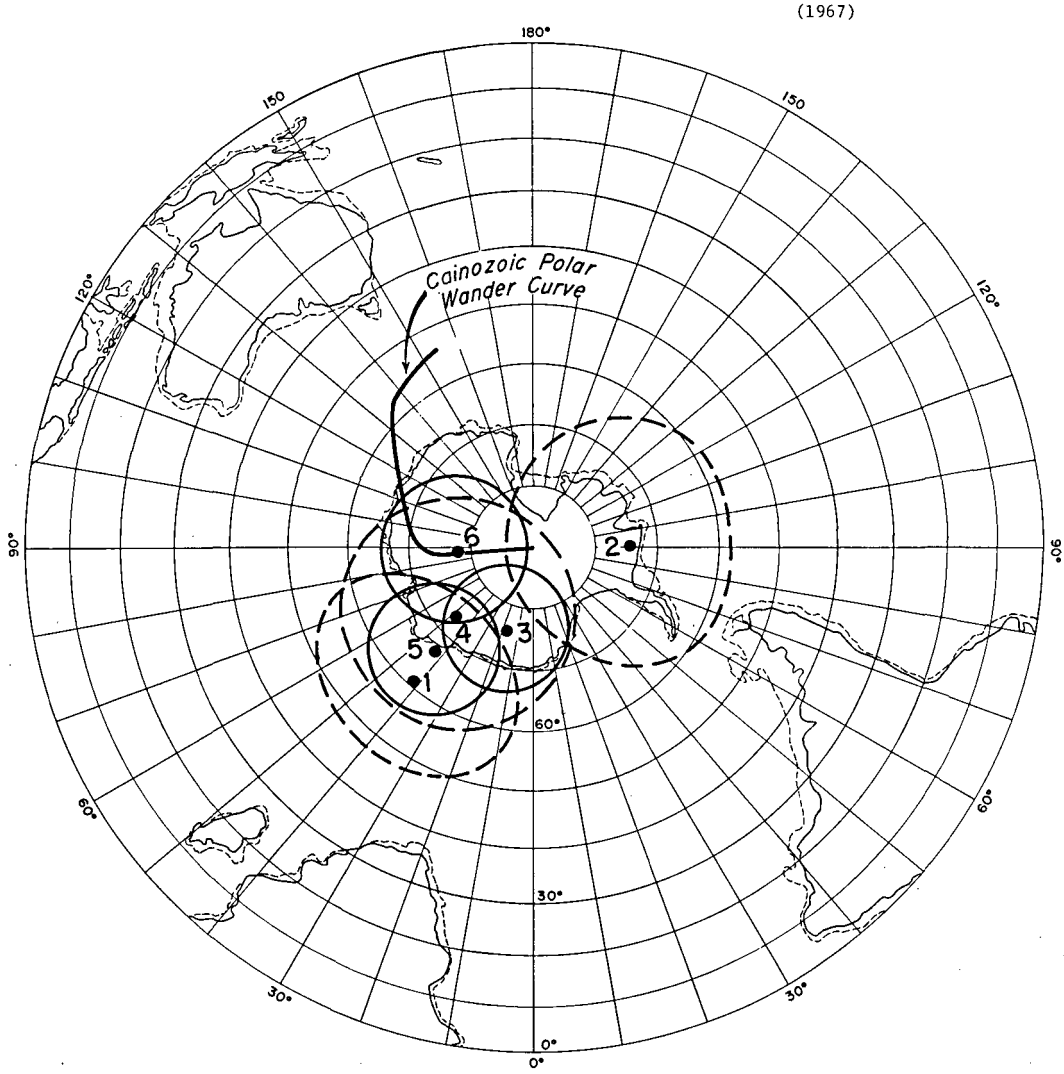


Fig. 6. Equal-area polar projection of the southern hemisphere showing poles given in Table 2. Pole 5 is the final Macquarie Island pole from this study and pole 6 is a pole of similar age from the Australian continent. Also shown is a Cainozoic polar wander curve from Australia obtained by Wellman, McElhinny & McDougall (1969) and McElhinny, Embleton & Wellman (1974).

namely, the average strike of the dykes in block 2 is east-northeasterly similar to the strike of the interpreted seafloor-spreading anomalies whereas the average strike of the dyke sites in block 3 is southeasterly. It is thus possible that block 3 may have a rotational component around a vertical axis relative to block 2. The simple bedding correction could not compensate for such a rotation.

To test the above possibility, the directions for sites in block 3 were rotated by 55° in an anticlockwise sense so that the average strike of the dykes in block 3 coincided with the average strike of the dykes in block 2. Both blocks would then be expected to be in their original orientation as defined by regional seafloor anomalies. This correction (Fig. 4c) produced a palaeomagnetic pole for block 3 which is consistent with that in block 2. A final average pole for Macquarie Island using the bedding corrected directions for block 2 and the rotated and bedding corrected directions for block 3 was then calculated at Lat. 68°S, Long. 43°E with a 95 percent circle of confidence of 12° radius. The various poles are summarised in Figure 6 and Table 2.

DISCUSSION

The effectiveness of the two corrections used in this study is consistent with the implied model for the generation of ocean crust. The success of the bedding correction in reducing the 95 percent confidence radius from 25° to 12° argues that the lavas were laid roughly horizontally and the dykes injected roughly vertically. The success of the rotation correction on the directions from block 3 which brought the palaeomagnetic poles from blocks 2 and 3 into statistical agreement, also argues that dykes are injected roughly parallel to the spreading centre (and to the consequent seafloor-spreading anomalies).

The residual dispersion on the directions of magnetization after correction could be the result of the lavas' being laid down subhorizontally and the dykes being injected subvertically and subparallel to the spreading centre. However, some workers (Lowrie, 1973, Ade-Hall & Kitazawa, 1973) have observed considerable variation in magnetic inclinations measured in submarine basalts, compared to the average inclination: this is possibly due to the effects of secular variation, or to hard components of viscous remanent magnetization as Lowrie (1973) claims. The mean difference (r.m.s.) in inclination described is 17° and, if such secondary magnetizations occur commonly in seafloor rocks, then a large dispersion may be normal when dealing with these rocks.

TABLE 3

Magnetic Properties (Average)

Rock Type	Number of Samples	NRM	σ_1	SUS	σ_2	K	MDF
basalt	22	1243.1	1300.	1637.5	780.	1.2	90
dolerite	49	231.9	230.	355.6	170	3.3	170
red-sediment	2	29.7		131.7		0.4	100
gabbro	1	1352.4		2638.0		0.75	150
serpentinites	3	1116.3		3300.		0.5	80

NRM = Natural remanent magnetization x 10^{-6} gauss
SUS = Susceptibility x 10^{-6} gauss/oe
K = Koenigsberger ratio (field = 65,000)
MDF = Median destructive field (oersteds)
$\sigma_{1,2}$ = Standard deviations of NRM and SUS respectively

Model studies by Williamson (1974) also demonstrate that the geographical distribution of stable sites having directions of magnetization with the same and opposite polarities to the Earth's present field is reflected in the magnetic anomalies on Macquarie Island. These are presumably seafloor-spreading anomalies.

The average magnetic properties obtained in the present study (Table 3) indicate that, although the major contribution to such anomalies comes from the lavas as a result of their higher NRM's, the remainder of the igneous rocks making up the oceanic crust (with the possible exception of serpentinites) appear to be palaeomagnetically stable. Moreover, it appears that the gabbros may have sufficiently high NRM's to contribute to seafloor-spreading anomalies also.

The interpretation by Williamson (1974) of marine magnetic anomalies around Macquarie Island indicates that the island rocks were formed at the Indian-Antarctic spreading ridge at about 27 m.y. ago. Consequently the pole from the island should correspond to the apparent pole for rocks on the Australian continent which have the same age. The poles obtained after bedding correction, and after bedding correction and rotation, are both consistent with the apparent pole for 22-25 m.y. ago obtained in Southeast Queensland on the Australian mainland by Robertson (1966) and Webb *et al.* (1967). The results of the present study are thus consistent with Macquarie Island rocks forming as part of the Indian plate. The pole obtained is, however, west of a perfect fit with the polar wander path of the Australian plate as calculated by Wellman, McElhinny & McDougall (1969) and McElhinny, Embleton & Wellman (1974). This difference could be caused by the small number of stable sites in the present study or by unaccounted-for structural effects. The latter

would not be surprising considering the structural complexity of the island.

ACKNOWLEDGEMENTS

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